

ZHE JIA, Ph.D.

AI Scientist / Research Engineer • AI4Science, LLMs & Foundation Models

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Professional Summary

Machine Learning Scientist and Principal Investigator architecting Foundation Models for Earth systems, PDE dynamics, and LLM pretraining. Caltech Ph.D. with 8+ years bridging computational physics, distributed HPC, and deep learning. Demonstrated expertise across the **full LLM lifecycle** (BPE, RoPE, GQA, FlashAttention, DDP, IsoFLOP scaling), **mechanistic interpretability** (causal probing of Neural Operators), and **rigorous model evaluation** on complex topologies. Author of 20+ top-tier papers (Science, NeurIPS, ICLR). Highly equipped for framework-level roles spanning weather/climate modeling, Large Language Models (LLMs), Embodied AI, and production-grade GPU-accelerated research engineering.

Technical Skills

ML Architectures:	LLMs, Graph/Set Transformers, Fourier Neural Operators (FNO), State Space Models, Latent Diffusion.
Pretraining & Systems:	PyTorch, CUDA (Graphs), FlashAttention, DDP, ZeRO, muP, KV-Cache Streaming, MinHash/LSH.
Physics & Data Sci:	Solid and Fluid Dynamics, PDE/ODE Solvers, Bayesian Inversion, Spectral Methods, Adjoint Methods.
Languages & Tools:	Python, C++, Fortran, Bash, Modular Software Design, Linux, HPC Slurm, CI/CD, Git.

Selected ML Systems & Foundation Models

nanoGPT-Seis: End-to-End LLM Pretraining Lifecycle & IsoFLOP Scaling

[Open-Source Project](#)

- **Full-Stack Pretraining:** Engineered an end-to-end 113M parameter LLM from scratch. Built web crawlers, implemented MinHash/LSH deduplication, trained a custom BPE tokenizer (16k vocab), and deployed DDP mixed-precision (bf16) training across NVIDIA A30 GPUs.
- **Modern Architecture:** Implemented state-of-the-art transformer components in pure PyTorch, including Grouped-Query Attention (GQA), Rotary Positional Embeddings (RoPE), RMSNorm, SwiGLU, and FlashAttention-2.
- **Scaling Laws & muP:** Conducted rigorous compute-optimal (IsoFLOP) scaling sweeps (3M–85M params) using Maximal Update Parametrization (muP) to achieve width-independent learning rate transfer.
- **Inference Engine:** Built a KV-cached streaming inference engine achieving 176ms time-to-first-token with anti-repetition sampling, demonstrating a 35% fluency boost via domain-to-general text mixing.

Mechanistic Interpretability for Neural Operators (PDE Surrogates)

[NeurIPS 2026 under review](#)

- **Causal Physical Probing:** Developed a cancellation-free Fourier coupling probe (mode ablation & activation patching) to dissect FNO computations. Proved FNOs learn off-diagonal coupling to exactly represent PDE advection terms.
- **System Precision Debugging:** Identified and resolved catastrophic cancellation bugs in `float32` spectral finite-differences, deploying `float64` autograd Jacobians to accurately extract high-frequency spectral signals.
- **Actionable Architectural Fix:** Discovered that FNOs systematically under-respond to transport (Péclet Law). Designed a mechanism-motivated spectral ($\dot{H}^1$) loss that repaired the coupling bias, improving long-horizon autoregressive rollout stability by **40%** on 2D Navier-Stokes systems.

Rigorous Evaluation Methodologies for Neural PDE Surrogates

[TMLR to be submitted](#)

- **Metric Design & Pathology Discovery:** Exposed a critical evaluation pitfall in SciML: proved that aggregate RMSE heavily rewards "blur" (collapsing dynamic fronts to static backgrounds) in sparse, localized physical fields.
- **Diagnostic Engineering:** Designed architecture-agnostic front-sensitive metrics and effective receptive-field diagnostics. Demonstrated that global attention networks degenerate into effective locality while failing to capture front-amplitude fidelity.
- **Structural Ablation:** Executed robust 2×2 grid analyses comparing local attention vs. message-passing (GNS) vs. global attention, proving local priors consistently maintain front-fidelity across varying datasets and capacities.

GTS: Graph Transformer Simulator for Ultra-Long Earth Dynamics

[ICLR submission in prep](#)

- **Continuous State Space Modeling:** Architected a Graph Transformer supporting **10,000+ step** autoregressive rollouts across physical dynamics spanning **15 orders of magnitude**. Directly targets autoregressive drift and memory-wall bottlenecks in AI weather/climate models.
- **Hardware Acceleration:** Suppressed compounding error via Truncated BPTT. Leveraged PyTorch 2.0 compiler and **CUDA Graph Capture** with physical adaptive time-stepping to achieve a **3x zero-loss forward GPU acceleration**.

Sensoformer: Foundation Model for Irregular Physical Sensor Sets

[NeurIPS 2026 under review](#)

- **Architecture Design:** Built a variable-cardinality SciML framework for multimodal sensor arrays utilizing Siamese 1D-ResNet waveform towers and masked Set-Transformer self-attention. Achieved $O(1)$ global routing avoiding GNN over-smoothing.
- **Sim-to-Real Alignment:** Designed *Physics-Structured Domain Randomization (PSDR)* as causal intervention over nuisance variables, achiev-

ing SOTA amortized inference on real-world distributions with zero real labels.

- **Systems Optimization:** Engineered dynamic tensor batching, double masking, and a continuous Focal L1 loss to cure long-tail mode collapse, delivering **1000x lower-latency** inference versus iterative scientific inversions.

Professional Experience

Assistant Research Professor / Principal Investigator
The University of Texas at Austin, Institute for Geophysics

2024 – Present
Austin, TX

- Lead an AI4Science lab translating complex physical inverse problems into scalable ML frameworks: Foundation Models for irregular sensor sets (Sensoformer), graph neural simulators, and uncertainty-aware generative physical AI.
- Own end-to-end model architecture, software design, and compute planning. Manage roadmap and mentor Ph.D./postdoc researchers across ML, HPC, and physics workflows.
- Secured **5M+ CPU/GPU-hours** in supercomputing allocations and engineered **10TB+** automated data pipelines for unstructured mesh and sensor datasets, optimizing I/O for variable-size sequence training.

Cecil H. and Ida M. Green Fellow / Technical Lead
Scripps Institution of Oceanography, UC San Diego

2022 – 2024
San Diego, CA

- Led a cross-institutional team combining nonlinear dynamic rupture simulation, large-scale geophysical inversion, and distributed HPC (OpenMPI) to resolve the complex 2023 Kahramanmaraş earthquake doublet (**Science Cover**).
- Engineered robust ETL systems for multi-TB, extreme low-SNR time-series data to identify cascade mechanisms.

Graduate Researcher (Scientific Computing & ML)
California Institute of Technology (Caltech)

2016 – 2022
Pasadena, CA

- Developed a continuous CNN-UNet deep learning framework for signal extraction, delivering a **100x speedup** over classical Digital Signal Processing (DSP) workflows and enabling scalable global data analysis.
- Built highly parallelized Bayesian inversion engines using hybrid MCMC and adjoint optimization for high-dimensional uncertainty quantification. Processed arrays with thousands of stations and multi-TB datasets.

Education

Ph.D. in Computational Geophysics (Inverse Problems & Optimization)
California Institute of Technology (Caltech)
B.S. & M.S. in Geophysics (Special Class for the Gifted Young - Top 1%)
University of Science and Tech. of China (USTC)

2022
Pasadena, CA
2013 / 2016
Hefei, China

Selected Publications

Selected from 23 peer-reviewed papers (over 1,200 citations) spanning ML and Science venues. (: corresponding author)*

- Jia, Z.*, (2026). *What Do Fourier Neural Operators Compute? A Causal Coupling Probe and a Péclet Law for Spectral Mixing*. **NeurIPS** (in review draft).
- Jia, Z.*, et al. (2026). *When aggregate RMSE rewards blur: evaluation pitfalls for neural mesh-PDE surrogates*. **TMLR** (to be submitted draft).
- Jia, Z.*, Zhang, X., Li, J. (2026). *Sensoformer: Robust Sim-to-Real Inference on Variable-Geometry Sensor Sets via Physics-Structured Randomization*. **NeurIPS** (in review). [[arXiv:2601.06320](https://arxiv.org/abs/2601.06320)]
- Liu, D., Jia, Z.*, et al. (2026). *GTS: A Graph Transformer Foundation Model for Ultra-Long Autoregressive Dynamics*. **ICLR** (in prep).
- Zhang, X., Jia, Z.*, Ross, Z., Clayton, R. (2020) *Extracting dispersion curves from ambient noise correlations using deep learning*. **IEEE TGRS**.
- Jia, Z.*, et al. (2023) *The complex dynamics of the 2023 Kahramanmaraş, Turkey, Mw 7.8-7.7 earthquake doublet*. **Science** (Cover Article). – Large-scale HPC multi-modal inversion (200+ citations).
- Ross, Z., Idini, B., Jia, Z., et al. (2019) *Hierarchical interlocked orthogonal faulting in the 2019 Ridgecrest earthquake sequence*. **Science**. – High-impact dynamic system analysis (500+ citations).

Selected Honors & Awards

Cecil H. and Ida M. Green Fellowship | \$138k independent award for research excellence
Outstanding Self-Financed Student Award | National Award, Top 1%
Caltech GPS Fellowship | Full graduate study fellowship
China National Scholarship | Top 1% of graduate students nationwide

2022 – 2024
2023
2016
2015